

Logistic Regression using Class Weights

Accuracy: 0.9985

	Precision	Recall	F1 Score
Not Fraud	1	1	1
Fraud	0.56	0.82	0.67

Logistic Regression using Undersampled Data:

Accuracy: 0.975

	Precision	Recall	F1 Score
Not Fraud	1	0.98	0.99
Fraud	0.06	0.88	0.11

Logistic Regression using SMOTE:

Accuracy: 0.9760

	Precision	Recall	F1 Score
Not Fraud	1	0.98	0.99
Fraud	0.06	0.86	0.11

Decision Tree on Test:

Accuracy: 0.9993

	Precision	Recall	F1 Score
Not Fraud	1	1	1
Fraud	0.86	0.71	0.78

Decision Tree on Train:

Accuracy: 0.9998

	Precision	Recall	F1 Score
Not Fraud	1	1	1
Fraud	0.99	0.92	0.95

KNN:

Accuracy: 0.9994

	Precision	Recall	F1 Score
Not Fraud	1	1	1
Fraud	0.91	0.70	0.79

Naive Bayes:

Accuracy: 0.978

	Precision	Recall	F1 Score
Not Fraud	1	0.98	0.99
Fraud	0.06	0.8	0.11

XG Boost on Test

Accuracy: 0.9994

F2-score: 0.7479

	Precision	Recall	F1 Score
Not Fraud	1.00	1.00	1.00
Fraud	0.95	0.71	0.81

XG Boost on Train

Accuracy: 0.9995

	Precision	Recall	F1 Score
Not Fraud	1.00	1.00	1.00
Fraud	0.95	0.77	0.85

XG Boost (tuned)

Accuracy: 1.00

F2-score: 0.8069

	Precision	Recall	F1 Score
Not Fraud	1.00	1.00	1.00
Fraud	0.88	0.79	0.83

XGBoost (tuned) — Train Results

Accuracy: 1.00

	Precision	Recall	F1 Score
Not Fraud	1.00	1.00	1.00
Fraud	1.00	1.00	1.00

Random Forest on Test

Accuracy: 0.9995

F2-score: 0.7887

	Precision	Recall	F1 Score
Not Fraud	1.00	1.00	1.00
Fraud	0.95	0.76	0.84

Random Forest on Train

Accuracy: 0.9996

	Precision	Recall	F1 Score
Not Fraud	1.00	1.00	1.00
Fraud	0.96	0.81	0.88

Random Forest Tuned

Accuracy: 1.00

F2-score: 0.8077

	Precision	Recall	F1 Score
Not Fraud	1.00	1.00	1.00

Fraud	0.77	0.82	0.79
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